

Energy-Based Interpretation of General Relativity

A Lay Hypothesis

Author: Andrew Robus

Date: October 9, 2026

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

1. Starting Point of GR

General Relativity describes gravitation geometrically. Matter and energy are described by the energy-momentum tensor $T_{\mu\nu}$. The Einstein field equations read:

- $G_{\mu\nu} = (8 * \pi * G / c^4) * T_{\mu\nu}$

Where:

- $G_{\mu\nu}$ describes the curvature of spacetime,
- $T_{\mu\nu}$ describes the local distribution of energy, momentum, pressure, and stresses.

The geometry of spacetime, in turn, determines the possible motion of energy and matter. This yields the familiar relationship:

- $T_{\mu\nu} \rightarrow g_{\mu\nu} \rightarrow \text{Dynamics} \rightarrow T_{\mu\nu}$

2. My Fundamental Modification

The hypothesis initially does not introduce a new theory of gravitation. Instead, the ontological separation between:

- matter,
- radiation,
- field energy,
- and other forms of energy

is abandoned at the fundamental level.

The core assumption is:

- $\text{Universe} = E$

The universe fundamentally consists of energy in different states, distributions, and forms of motion. In this interpretation, "matter" is not an independent building block.

- $\text{Matter} = \text{bound energy state}$

Radiation, on the other hand, is a largely free energy state.

3. What Is Bound Energy?

In this hypothesis, matter arises when several photons permanently combine into a shared, stable structure. The photons do not simply lie side by side, but can overlap significantly and jointly form a compact, slightly irregular shape—conceived in simplified terms as a small, dynamic sphere.

Within this structure, the photons move and change together around a shared center of energy and gravity without losing their strong spatial overlap. In this way, a quark, for example, could be understood as such a bound photonic structure.

Which shape such a structure takes and whether it remains stable likely depends on several properties of the participating photons—particularly their energy, polarization, spin, and mutual interactions. The properties of the photons do not necessarily have to be identical. Certain combinations or even opposing orientations could be crucial for stability.

Gravity plays an important role here, as the shared energy distribution itself could contribute back to the binding of the structure.

It is experimentally and theoretically known that photons can form bound collective states under suitable conditions. The hypothesis asks the further question of whether such photonic organizational forms—under the extreme conditions of the early universe (before, during, or after the Big Bang) and taking gravity into account—could form a fundamental basis of stable matter.

Initial Rough Calculation – Order of Magnitude Only

To initially get a sense of scale, a very simple sample calculation can be performed. It is by no means intended to prove that a quark consists of photons, but merely to show that theoretically, a large number of photons would not necessarily be required.

As an example, the mass of an up quark of approximately $m_u \approx 2.16 \text{ MeV} / c^2$ is used. The corresponding rest energy is thus approximately $E_u \approx 2.16 \text{ MeV}$.

If we now hypothetically take photons each with $E_{\text{gamma}} \approx 0.72 \text{ MeV}$, we already get:

- $3 * 0.72 \text{ MeV} = 2.16 \text{ MeV}$

Purely in terms of energy, three photons would already reach the scale of the assumed rest energy of an up quark.

This is explicitly only an initial rough calculation. In the hypothesis, the three photons would need to form a collective bound structure and be arranged such that their total energy and total momentum yield the properties of the assumed particle. Spin, polarization, spatial overlap, interactions, and potential gravitational binding are not yet accounted for in this simple calculation.

The calculation therefore merely shows: If the hypothesis were fundamentally possible, such an elementary structure would not necessarily have to consist of a very large number of photons. A few sufficiently energetic photons could, purely energetically, reach the order of magnitude of a quark's rest energy.

This is not proof of a photonic structure of quarks, but merely a first quantitative illustration of the order of magnitude.

4. The Energy-Momentum Tensor Remains Intact

The crucial point is: We do not replace $T_{\mu\nu}$. On the contrary: It remains the central mathematical tool for describing energy and momentum distribution.

- $T_{\mu\nu}$ = local energy, momentum, and stress configuration

It continues to contain, in particular:

- energy density,
- momentum density,
- energy flux,
- pressure,
- stresses.

Thus, no new "matter equation" needs to be introduced for matter. A matter system continues to be described by its corresponding $T_{\mu\nu}$.

Our modification consists of stating that matter is ontologically not an additional basic component. It is a bound, organized state of the underlying energy—in this hypothesis, presumably an organized structure of photons.

The energy and momentum distribution of this bound structure continues to be described by the energy-momentum tensor.

5. The Einstein Field Equation Remains Unchanged

Consequently:

- $G_{\mu\nu} = (8 \cdot \pi \cdot G / c^4) \cdot T_{\mu\nu}$

remains fully intact. The change on the right-hand side is not a new formula, but a change in its interpretation.

- **Standard interpretation (simplified):** Matter / Energy $\rightarrow T_{\mu\nu}$
- **Our interpretation:** Energy $\rightarrow T_{\mu\nu}$

Matter is merely a special state of this energy.

6. Energy and Geometry

Within the hypothesis, the following chain emerges:

- $E \rightarrow T_{\mu\nu} \rightarrow G_{\mu\nu} \rightarrow g_{\mu\nu}$

Where $G_{\mu\nu}$ describes spacetime curvature, while $g_{\mu\nu}$ describes the underlying spacetime geometry (metric).

Energy thus determines spacetime geometry via its energy-momentum state. Conversely, geometry determines the possible development of energy and its states of motion.

Thus, the mutual feedback loop of GR remains intact:

- $T_{\mu\nu} \leftrightarrow g_{\mu\nu}$

7. The Universe as a Single Total System

The hypothesis further assumes that there is no external space and no external medium into which the universe expands. Instead, the universe is understood as a single connected physical total system.

The initial state is assumed to be an extremely concentrated and high-energy sphere. This became unstable and expanded explosion-like. In this picture, the universe would be the expanding and cooling structure of this original explosion.

The interior space created in the process is understood within the hypothesis as an electromagnetic field. Its interaction with the energy contained within it is to be investigated within the framework of the Einstein-Maxwell equations.

Put simply, matter and other bound structures would be the cooled and organized components of this original high-energy evolution.

To be investigated: Does the Hubble constant emerge "automatically" in this system?

8. What Actually Changes in GR?

This can be stated very concisely:

Unchanged:

- The field equation $G_{\mu\nu} = (8 \cdot \pi \cdot G / c^4) \cdot T_{\mu\nu}$ remains intact.
- The geometric description of gravity, covariant energy-momentum conservation, and geodesic motion are also initially retained.

Changed Ontology:

- Instead of viewing matter, radiation, and fields as distinct fundamental physical contents, the hypothesis states: Everything is an expression of energy.
- Through the properties and interactions of individual photons—such as polarization, gravity, and further properties yet to be specified—stable structures could form from which elementary particles like quarks and other forms of matter might emerge.
- Matter is understood as a stable or long-lived, organized state of this energy or these photonic structures.
- The various physical states are described regarding their gravitationally relevant properties by their respective energy-momentum tensor $T_{\mu\nu}$.

9. How Does Special Relativity Fit In Here?

Since within this hypothesis we are an expanding "explosion," our expansion—and thus the development of our space—is directly linked to time. The longer ago the Big Bang occurred, the larger the resulting interior space. Conversely, a larger interior space means a more advanced temporal evolution. This is why we speak of spacetime.

The fundamental energy propagates in the form of photons at the speed of light c . Its propagation forms an essential part of the spatial evolution of the system. The energy distribution and its gravitational effect, together with spacetime geometry, determine further evolution. The interaction

between the electromagnetic field and spacetime could be described within the framework of the Einstein-Maxwell equations.

As expansion progresses, energy spreads over an increasingly large space and cools down. Time, entropy, and the observed evolution of the universe would ultimately be different aspects of the same ongoing process: the expansion and cooling of the original explosion.

Special Relativity specifically describes how space and time relate to each other for moving energy and observers. Whether its known mathematical relations can be fully derived from this model remains an open question.

10. Info

Although this hypothesis considers photons as possible fundamental building blocks of all matter, it is not ruled out that quarks and other elementary particles are based on other forms of energy particles or energy states. The fundamental assumption "Universe = E" remains unaffected by this.

11. Speculative Additional Hypothesis on the Origin

The hypothesis so far primarily describes the physical evolution of our universe. A further, significantly more speculative assumption concerns its origin.

It is possible that our universe itself emerged from the collapse of an earlier gravitational system. Following the formation of an event horizon, its internal evolution could have decoupled completely from the original universe. The collapsed system would then not simply be an object in a shared external space, but could represent the starting point of its own universe.

In this model, an extreme concentration of energy—for example, through energy released or redistributed during gravitational contraction, comparable to the physical processes in the contraction of compact astrophysical objects—would ultimately lead to an unstable state from which the expansion of our universe emerged.

Acknowledgment of AI Assistance

Artificial intelligence (ChatGPT) was used as an assisting tool during the development of this hypothesis, including discussions of the underlying concepts, language formulation, mathematical notation and equations, and the research and verification of relevant information and sources.

However, the fundamental idea, conceptual approach, underlying assumptions, and overall direction of the hypothesis are my own. The AI served as an aid in articulating, discussing, and refining my ideas, rather than as the originator of the hypothesis.

The author takes sole responsibility for the content, assumptions, interpretations, and conclusions presented in this work.